BENCHMARK 2

Proof Transfer and Completion Diagnostic R1.0

Candidate identifier: _____ **Date:** _____

This is a second formative assessment, not an official BMO or IMO paper. It is designed to check whether mathematical methods learnt in the study notes can be used independently on *new* problems. It is not an intelligence test or an estimate of an IMO rank. **How to sit the assessment**

1. The four subjects are separate sittings. Preferably complete them across two to four days. Rest between sittings. It is not necessary to finish all eight questions tomorrow.
2. Write reasons, not just answers. An unfinished but valid argument can earn credit. If you are unsure, say what you can prove and what remains open.
3. No notes, hints, calculator, internet or AI during a timed sitting. After stopping, further exploration is welcome but must be labelled *after time*.
4. Record approximately how many minutes you spent on each question, whether you had seen it or an equivalent theorem before, and whether you used any support. These are diagnostic details, not penalties.
5. A photograph that is not supplied is *missing evidence*, not automatically an unsuccessful attempt. Marks should only be assigned to work actually available for review.
6. Participation, pauses and sharing of work remain your choice. Marks need not be disclosed if you prefer feedback without numbers.

Subject	**Time allowed**	**Maximum marks**
Algebra	45 minutes	14
Geometry	55 minutes	14
Number Theory	45 minutes	14
Combinatorics	45 minutes	14
Total	**190 minutes**	**56**

For the mentor. Do not compare raw scores directly with Benchmark 1 without noting different question content, possible previous exposure, changed instructions and timing uncertainty. The private marking pack contains rubrics and audit instructions. Keep it separate from this student booklet.

Algebra

Independent diagnostic, two proof questions

Time allowed: 45 minutes **Total: 14 marks**

Candidate identifier: _____ **Date:** _____

Instructions. Attempt both questions independently. Show complete arguments, including all relevant cases and any final verification. No internet, calculator, AI tools or hints during the timed sitting. If unfinished, write down any useful progress. Stop the timed work when time is called.

Question 1 **[7 marks]**

Let $f : \mathbb{Z} \to \mathbb{Z}$ satisfy

$$f(x+y) + f(x-y) = 2f(x) + 2y^2$$

for all integers x, y. Find every such function f. Prove both that your list is complete and that every function on your list satisfies the equation.

Question 2 **[7 marks]**

Let a, b, c be positive real numbers. Prove that

$$\frac{a^2}{a+b} + \frac{b^2}{b+c} + \frac{c^2}{c+a} \geq \frac{a+b+c}{2}.$$

Determine exactly when equality holds.

After the sitting (not part of the timed assessment): Record approximate time spent on each question, whether either problem was familiar, and whether you worked beyond the limit. If you need more space, use the separate answer sheets.

Q1: _____ min Q2: _____ min Prior exposure: _____

Geometry

Independent diagnostic, two proof questions

Time allowed: 55 minutes **Total: 14 marks**

Candidate identifier: _____ **Date:** _____

Instructions. Attempt both questions independently. Show complete arguments, including all relevant cases and any final verification. No internet, calculator, AI tools or hints during the timed sitting. If unfinished, write down any useful progress. Stop the timed work when time is called.

Question 1 **[7 marks]**

In an acute triangle ABC, let D, E and F be the feet of the perpendiculars from A, B and C to BC, CA and AB, respectively. Prove that

$$\angle EDF = 180^\circ - 2\angle BAC.$$

A clear labelled diagram may be used, but all claimed angle relationships must be justified.

Question 2 **[7 marks]**

Points A, B, C, D lie on a circle in that order, and form a convex quadrilateral. Without quoting Ptolemy's theorem, prove that

$$AC \cdot BD = AB \cdot CD + BC \cdot AD.$$

You may introduce auxiliary points or lines. Justify any triangles you claim are similar.

After the sitting (not part of the timed assessment): Record approximate time spent on each question, whether either problem was familiar, and whether you worked beyond the limit. If you need more space, use the separate answer sheets.

Q1: _____ min Q2: _____ min Prior exposure: _____

Number Theory

Independent diagnostic, two proof questions

Time allowed: 45 minutes **Total: 14 marks**

Candidate identifier: _____ **Date:** _____

Instructions. Attempt both questions independently. Show complete arguments, including all relevant cases and any final verification. No internet, calculator, AI tools or hints during the timed sitting. If unfinished, write down any useful progress. Stop the timed work when time is called.

Question 1 **[7 marks]**

Prove that 30 divides $n^5 - n$ for every integer n. Your proof should explain the divisibility by each of the primes 2, 3 and 5.

Question 2 **[7 marks]**

For each positive integer n, determine

$$\gcd\bigl(n^2 + 1, (n+1)^2 + 1\bigr).$$

Give an exact answer depending on n, and prove that no other values of this greatest common divisor are possible.

After the sitting (not part of the timed assessment): Record approximate time spent on each question, whether either problem was familiar, and whether you worked beyond the limit. If you need more space, use the separate answer sheets.

Q1: _____ min Q2: _____ min Prior exposure: _____

Combinatorics

Independent diagnostic, two proof questions

Time allowed: 45 minutes **Total: 14 marks**

Candidate identifier: _____ **Date:** _____

Instructions. Attempt both questions independently. Show complete arguments, including all relevant cases and any final verification. No internet, calculator, AI tools or hints during the timed sitting. If unfinished, write down any useful progress. Stop the timed work when time is called.

Question 1 **[7 marks]**

In a competition with $n \geq 1$ players, each pair of distinct players has played exactly one match, with no draws. Prove that the players can be listed in an order $P_1, P_2, \ldots, P_n$ such that P_i defeated P_{i+1} for each $1 \leq i < n$.

Question 2 **[7 marks]**

A $3 \times n$ rectangular grid of unit squares is coloured using red and blue. A *monochromatic rectangle* consists of four squares at the intersections of two distinct rows and two distinct columns, all of the same colour. Find the smallest positive integer n such that *every* red-blue colouring of a $3 \times n$ grid contains a monochromatic rectangle. Prove both that your number always works and that one fewer column may fail.

After the sitting (not part of the timed assessment): Record approximate time spent on each question, whether either problem was familiar, and whether you worked beyond the limit. If you need more space, use the separate answer sheets.

Q1: _____ min Q2: _____ min Prior exposure: _____